# 6. The Data Layer Reforged

## Storage, semantics, and reasoning in the converged data substrate

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# Executive Summary

For decades, the data industry has been organized around a 2x2 matrix: structured versus unstructured data on one axis, transactional (OLTP) versus analytical (OLAP) workloads on the other. Each quadrant developed its own champion, its own architecture, and its own economics. The history of the industry since 2020 is the history of this matrix collapsing — first along the structured/unstructured axis within analytics (the lakehouse), and now, since 2025, along the transactional/analytical axis (the Postgres acquisition wave and the rise of the "lakebase" category). The agent of this collapse, literally, is the AI agent: a new class of data consumer that reads and writes at machine speed, dissolving the human-paced rhythms around which the old fragmentation was tolerable.

## Key Theses at a Glance

This report advances five theses.

**Thesis 1 — The two convergences are demand-driven, not vendor-driven.** The lakehouse unified storage because AI workloads begin with unstructured data; the OLTP–OLAP reunification is happening because agents must analyze and act in a single loop. Critically, the second convergence is proceeding by *composition* (dedicated engines unified at the platform level) rather than by the single-engine HTAP fusion the industry expected a decade ago. HTAP engines found real commercial niches — SAP HANA most prominently — but never became the market's default architecture; among today's converged-market contenders, Oracle is the most aggressive large incumbent still pursuing the fusion path, and its trajectory is the instructive counter-case.

**Thesis 2 — Storage is commoditized; value is migrating upward through two new strata.** Open table formats (Apache Iceberg, Delta Lake) neutralized storage as a moat. Value is accruing first to the *context layer* — catalogs, semantic models, and ontologies that tell agents what data means — and second to an *analytical reasoning layer* in which prediction, causal inference, and what-if simulation become high-frequency, agent-executed workloads rather than scarce human projects. A modality lens sharpens the migration: LLMs made the enterprise's *qualitative* corpus — the majority of corporate information, stored as language rather than numbers — broadly and economically computable at a scale and flexibility no prior generation of NLP approached, arguably the largest revaluation of a dormant corporate asset class in the history of enterprise IT.

**Thesis 3 — The semantic layer is the new lock-in surface.** As data formats open, the proprietary residue concentrates in business meaning: metric definitions, entity models, causal structure. Platform vendors (Databricks, Snowflake, Google), independents (the merged Fivetran + dbt Labs, knowledge-graph platforms), and ontology maximalists (Palantir) are converging on this territory from different directions, and "semantic lock-in" is emerging as the successor to data lock-in. Against it runs a bottom-up developer counter-current — *compiled context*: LLM-maintained Markdown knowledge bases, formalized by Google's Open Knowledge Format (OKF) in June 2026 — which is simultaneously the lock-in's antithesis and, plausibly, its next distribution format.

**Thesis 4 — The analytical reasoning layer is the most underappreciated consumption driver.** Agents collapse the cost of advanced analytics from data-scientist-weeks to agent-

minutes, triggering a Jevons-style demand explosion, while their need to *act* forces analytics up the causal ladder from observation to intervention to counterfactual. One agentic decision can imply dozens of predictions and simulations — a structural compute multiplier for consumption-priced platforms, and a structural argument for new data primitives (branching, intervention logs, decision provenance).

**Thesis 5 — The endgame question is integration premium versus protocol federation.** Every contender's pitch is that data, semantics, and reasoning must live in one governed platform. Open protocols (MCP) and open semantic standards (Agents Schema, open-sourced Unity Catalog Business Semantics) are a standing counter-bet that context can become portable. Which force wins determines whether the converged data platform is the decade's great consolidation trade or a way station to re-fragmentation.

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# Part I. The Great Convergence

## 1. The Classical Data Layer: A 2x2 World

### 1.1 The four quadrants

The pre-convergence data layer can be mapped onto two axes. The first distinguishes *structured* data — rows and columns with enforced schemas — from *unstructured* data: documents, logs, images, audio, free text. The second distinguishes *transactional* workloads (OLTP), which manage the live state of applications through low-latency, ACID-guaranteed reads and writes, from *analytical* workloads (OLAP), which scan and aggregate large historical datasets to answer questions.

The four quadrants this produces were, for most of the industry's history, four separate markets with separate physics. Transactional systems are row-oriented and latency-optimized; analytical systems are columnar and throughput-optimized. Structured systems enforce schemas on write; unstructured stores defer schema to read time, or never impose one at all. These engineering trade-offs were genuine, and they justified separate engines, separate vendors, and separate buyer constituencies for decades.

### 1.2 Quadrant champions and their economics

Each quadrant produced a champion whose business model reflected its physics.

*Structured* × *analytical* was Teradata's market before it became Snowflake's. Snowflake's cloud-native separation of storage and compute, combined with consumption pricing, made it the defining franchise of the 2010s data stack, alongside hyperscaler equivalents (BigQuery, Redshift).

*Unstructured* × *analytical* passed from the Hadoop ecosystem to cloud object storage and then to Databricks, whose Spark lineage made it the default engine for data engineering and machine learning over raw, schema-light data.

*Structured* × *transactional* belonged to Oracle and SQL Server at the proprietary end and to PostgreSQL and MySQL in open source — with PostgreSQL, in particular, compounding quietly

into the most important open-source asset in the industry, a fact whose strategic significance only became visible in 2025 (Chapter 3).

*Unstructured* × *transactional* was the youngest quadrant: MongoDB's document model for flexible operational data, flanked by Elasticsearch for search and Redis for low-latency state.

### 1.3 The integration tax

The cost of this fragmentation was the pipeline. Because each quadrant held a partial copy of enterprise reality, data had to be perpetually extracted, transformed, and loaded across quadrant boundaries — OLTP systems replicated into warehouses overnight, lakes hydrated from application exhaust, operational stores synchronized via change-data-capture. An entire middleware industry (Informatica, Fivetran, dbt, CDC tooling) monetized the crossings. The integration tax had three components: *latency* (analytical truth lagged operational truth by hours or days), *duplication* (the same fact stored three to five times, each copy drifting), and *governance discontinuity* (access policies enforced per-system, breaking at every boundary). All three components were tolerable when the final consumer of analytics was a human reading a dashboard on a weekly cadence. None of them is tolerable when the final consumer is an agent acting in a loop measured in seconds — which is the subject of Chapter 4.

### 1.4 The orthogonal axis: modality, and the revaluation of qualitative data

The two axes of Section 1.1 — form and workload — explain the industry's historical structure: who sold which engine to whom. A third axis, orthogonal to the form axis, explains where value is moving now: **content modality**. Enterprise information divides into *quantitative data* — information reducible to numbers and codes — and *qualitative data* — meaning that resists such reduction: the argument inside a contract clause, the reasoning in a strategy memo, the frustration in a support ticket. (Multimedia — audio, image, video — forms a third class, addressed below.)

The distinction is best drawn functionally, by *processor* rather than by surface form. Quantitative data is whatever deterministic logic — code and SQL — can compute directly. On this criterion, categorical values belong to the quantitative class: a dimension table's product categories or status flags reduce losslessly to enumerable codes that software can group, join, and count, which is why the entire warehouse tradition handles them fluently. Qualitative data is what does *not* reduce — information whose relevant meaning survives only in natural language. Pre-LLM NLP could extract narrow, predefined signals from this class — sentiment scores, named entities, document classes, topic clusters — but each extraction required a task-specific model, schema, or pipeline; the meaning itself, in its full generality, remained inaccessible to computation.

The reason this axis matters becomes clear when it is placed against the anatomy of software itself. Software consists of three elements — interface, logic, and data — and the agentic era splits the logic layer in two: *deterministic logic*, expressed as code, and *probabilistic logic*, expressed as the LLM. The data layer splits along exactly the same line, and the two splits correspond: **code is the natural processor of quantitative data; the LLM is the natural processor of qualitative data.** Each half of the logic layer has its native half of the data layer. This correspondence is the simplest way to state what changed in 2023.

Stated as business history rather than architecture: for fifty years, the enterprise could compute its numbers systematically while extracting only narrow, predefined signals from

its own language. The numbers were refined, warehoused, governed, and dashboarded — the whole apparatus of Part I is the quantitative tradition's machinery. The language — contracts, memos, decks, email, meeting notes, the Word, PDF, and PowerPoint files in which most institutional knowledge actually lives — was mostly just *stored*, accumulating in shared drives and document systems as "dark data," commonly estimated at the substantial majority of enterprise information, valuable in principle and inert in practice; the sentiment engines and entity extractors that touched it skimmed predefined fields off its surface without making its meaning generally computable. What the LLM changed was the economics and the breadth: a single general-purpose interface can now interpret, summarize, compare, and synthesize heterogeneous language at enterprise scale, without a bespoke model per task. That shift made the bulk of the qualitative corpus economically and interactively computable — and it is difficult to identify a larger single revaluation of a dormant corporate asset class in the history of enterprise IT. The asset appears on no balance sheet.

Three consequences organize the rest of this report. First, the context layer examined in Chapter 5 is best understood as the *industrialization of the qualitative corpus*: RAG, vector indexes, and GraphRAG are its first-, second-, and third-generation process technologies, and the newest pattern — compiled context (Section 5.6) — is its fourth. Second, the modality axis adds resolution to the competitive map: the hyperscaler contest of Chapter 9 is in part a contest between modality fiefdoms, with Microsoft holding the world's largest governed qualitative corpus (M365 documents, mail, and meetings) while the data platforms hold the quantitative corpus — each side now racing to annex the other's modality. Third, multimedia is the revaluation-in-waiting: as multimodal models mature, meeting recordings, field imagery, and video move from storage cost to computable asset, replaying the qualitative revaluation on a second corpus. We flag it as a watch item rather than a load-bearing thesis.

## 2. The First Convergence: Lakehouse and the Commoditization of Storage (2020–2024)

### 2.1 Warehouse + lake unification

The first axis to collapse was structured versus unstructured, and it collapsed inside the analytical half of the matrix. Data warehouses offered performance, governance, and cost; data lakes offered cheap, schema-free storage at the price of disorder — the "data swamp" failure mode was a governance failure, not a storage failure. The lakehouse, pioneered commercially by Databricks on Delta Lake, resolved the dichotomy by layering warehouse properties (ACID transactions, schema enforcement, performance optimization) directly onto object storage. By 2024 the lakehouse was no longer a vendor differentiator but the default reference architecture, adopted in substance by every major platform including Snowflake. The cause was demand-side before it was vendor-side, per Thesis 1: machine-learning and then generative-AI workloads begin with raw, unstructured data, and enterprises needed warehouse-grade governance over precisely the assets the warehouse could not hold — the same demand logic that would later drive the second convergence (Chapter 4).

### 2.2 Open table formats and the storage–compute decoupling

The structurally decisive event inside the lakehouse transition was the victory of open table formats — Apache Iceberg and Delta Lake — which put transactional semantics and schema management on neutral object storage. The consequences ran deeper than file formats. Once

tables became engine-agnostic, the storage layer ceased to be a point of lock-in: any compliant engine could read, and increasingly write, the same governed tables. Snowflake's embrace of Iceberg and Databricks' acquisition of Tabular (founded by Iceberg's creators) were the two clearest signals that both leaders had reached the same conclusion — the moat was no longer where the bytes lived.

### 2.3 Strategic consequence: value migrates to catalog, governance, and compute

Commoditization at one layer relocates differentiation to the adjacent layer. With storage neutralized, competition moved to (a) the *catalog* — the system that knows what tables exist, what they mean, and who may touch them; (b) *governance* — lineage, quality, access policy, increasingly spanning structured and unstructured assets alike; and (c) the *compute engines* that monetize queries against open storage. Databricks' June 2025 Unity Catalog release made the logic explicit: full Iceberg REST Catalog support positioned Unity Catalog as the catalog through which external engines read and write governed, performance-optimized tables in either format, with the press materials framing the move as eliminating format lock-in while extending the catalog upward toward business users and metric definitions [1]. The catalog, in other words, was being groomed as the control point precisely because the storage beneath it had been given away. This pattern — open the lower layer, monetize the layer above — recurs throughout this report and is the single most reliable strategic playbook of the convergence era.

## 3. The Second Convergence: OLTP–OLAP Reunification (2025– )

### 3.1 Why single-engine HTAP stayed a niche; composition over fusion

The second axis began collapsing in 2025, and it collapsed in a way the industry did not predict. The 2010s vision was HTAP — hybrid transactional/analytical processing, one engine serving both workloads. That vision did not fail outright: SAP HANA achieved meaningful commercial adoption processing OLTP and OLAP in a single in-memory system, and platforms such as SingleStore demonstrated that combined transactional and analytical processing was viable for selected real-time workloads. What HTAP failed to become was the *default architecture* of the broader cloud data market — as of 2026 the dominant pattern remains a dedicated OLTP engine paired with a dedicated columnar engine. Both Snowflake and Databricks concluded that extending an OLAP engine into OLTP does not work, and chose instead to acquire production-proven PostgreSQL technology and *compose* a dedicated transactional engine into the platform — unification at the platform level (shared governance, shared catalog, shared developer surface) rather than the engine level [2].

### 3.2 The Postgres acquisition wave

The composition strategy expressed itself as an acquisition wave centered on one open-source project. Databricks acquired Neon (May 2025) and shipped Lakebase — a serverless, compute-storage-separated Postgres with copy-on-write branching, using object storage as the durable source of truth — reaching general availability in February 2026 and formalizing a new category label: the lakebase, a transactional layer native to the lakehouse [2]. Snowflake acquired Crunchy Data and productized it as Snowflake Postgres. Microsoft built rather than bought: Azure HorizonDB, now in public preview, is the most architecturally aggressive of the three, claiming scale-out to more than three thousand vCores, databases to 128TB, and roughly triple the transactional throughput of self-managed Postgres on a cloud-native storage layer [5].

That three platforms independently converged on Postgres-compatibility as the transactional substrate tells us two things: the structured × transactional quadrant's future volume is being contested at the Postgres commercialization layer rather than at the legacy-engine layer, and the quadrant's incumbent — Oracle — is being flanked rather than assaulted.

### 3.3 Oracle's counter-thesis: single-engine convergence

Oracle is the most aggressive large incumbent still prosecuting the fusion strategy — not the only vendor ever to attempt it, but the one whose 2026 release makes the bet most explicit. Oracle AI Database 26ai converges vector, JSON, graph, and relational data inside a single transactional engine, embeds persistent agent memory in the engine itself (the Unified Memory Core), and pushes security enforcement down to row, column, and cell granularity tied to both human and agent identities — an architecture marketed as eliminating the integration tax of fragmented AI stacks and recasting the database as the control point for enterprise automation [3][4]. The strategic argument is coherent: if agents act on data, the most governed place to host the action is inside the engine that owns the transaction. The structural weakness is equally clear: Oracle's gravity is strongest over existing systems of record, while agent-native greenfield workloads — the fastest-compounding demand pool, per Section 4.3 — default to Postgres-compatible and developer-led surfaces where Oracle's distribution is weakest.

## 4. The Catalyst: Agents as the Data Layer's First-Class Customer

### 4.1 From human-paced consumption to machine-speed read/write loops

Every architecture embeds an assumption about its consumer. The classical data layer assumed two consumers with different clocks: applications (and through them, end users) on the transactional side, operating in seconds; and human analysts on the analytical side, operating in days. The separation of OLTP and OLAP was tolerable because the analytical consumer was slow — a dashboard refreshed overnight was, for human purposes, fresh.

Agents break the assumption because they collapse the two consumers into one. An agent handling a supply-chain exception analyzes history (an OLAP act), decides, executes (an OLTP act), records the outcome (OLTP), and learns from it (OLAP) — in one loop, at machine speed, thousands of times per day. Modern agentic systems therefore demand low-latency writes, immediate analytical reads, and uniform governance simultaneously; keeping the two workload classes in isolated stacks adds latency, cost, and governance seams to every iteration of every loop. This is the demand-side cause of the second convergence, and it explains why the convergence began precisely when production agent deployments did.

### 4.2 New workload taxonomy: what agents store

Agents are not only a new consumer of existing data; they are a new producer of data categories that barely existed in 2023:

- **Agent memory** — episodic memory (interaction histories) with OLTP-like access patterns, and semantic memory (distilled facts and preferences) with knowledge-graph-like structure. Vendors are racing to internalize it: Oracle embeds agent memory in the database engine [4]; AWS ships it as a managed AgentCore capability [42].
- **Session state and checkpoints** — the live working state of long-running agents, requiring durable, low-latency transactional storage.

- **Branches** — agents experimentally fork database state, test, and discard. Neon reported before its acquisition that the majority of new databases on its platform were being created by agents rather than humans; copy-on-write branching is the primitive that makes this affordable, and it is foregrounded in Lakebase's design [2]. Branching recurs as a load-bearing primitive in the simulation discussion (Section 6.5).
- **Synthetic and derived data** — embeddings, extracted entities, evaluation sets, simulation outputs — produced at machine scale as a byproduct of agent operation.

The composite picture: the agent itself is a new OLTP demand source layered on top of its role as a new OLAP demand source.

### 4.3 Demand structure shift: stock versus agent-native greenfield

For competitive analysis it is essential to separate two demand pools. The *stock* — existing systems of record and their analytical shadows — moves slowly, defended by switching costs; Oracle's franchise lives here. The *flow* — agent-native greenfield workloads enumerated above — starts from zero and compounds at machine speed, because agents do not sleep, and every increase in agent capability increases their data appetite. The defining asymmetry of the next five years is that the flow is being won by different vendors than those who own the stock: serverless Postgres surfaces, document stores, and lakehouse-integrated transactional layers, rather than legacy engines. Market positions that look stable on a stock view look contested on a flow view.

### 4.4 The new ELT: semantic transformation

The pipeline layer is being transformed rather than eliminated. Classical ELT produced clean tables; agentic pipelines append three further stages — *embedding* (chunking and vectorizing unstructured content for retrieval), *entity/relationship extraction* (populating knowledge graphs and semantic models from documents, logs, and application exhaust), and *knowledge compilation* (LLM re-authoring of raw documents into curated, cross-linked, agent-readable knowledge bases — Section 5.6), whose output lands in a versioned file corpus rather than an index. The "T" in ELT is expanding from SQL transformation to semantic transformation, and the expansion is computationally expensive in a specific way: it consumes LLM inference. This is doubly favorable to the platform vendors — pipeline compute demand rises, and the governance argument for running that pipeline inside the platform (so that extracted semantics inherit catalog policy automatically) strengthens lock-in. The merged Fivetran + dbt Labs entity is positioning for exactly this expanded pipeline, framing itself as the infrastructure that makes agents trustworthy from data movement through transformation to the context an agent reasons from [16].

### 4.5 Agent economics on consumption models: the agent FinOps problem

Agent queries do not resemble dashboard queries. A dashboard executes a known query on a schedule; an agent investigating a question may issue dozens of exploratory queries, each consuming metered compute, before converging on an answer. For consumption-priced platforms this is a powerful revenue tailwind — and for customers it is a new and largely unsolved cost-control problem, because budget-based query routing, per-session spend ceilings, and cost-aware planning for agent workloads remain immature across the major platforms. We expect agent FinOps to follow the trajectory cloud FinOps followed a decade ago: an initial period of uncontrolled spend, a backlash, then a tooling category. The investment-relevant

nuance is timing asymmetry — the revenue tailwind arrives before the cost-control countermeasures do, flattering near-term consumption growth at Snowflake, Databricks, and the hyperscalers, while planting the seed of a later optimization headwind of the kind Snowflake already experienced once in the 2022–2023 warehouse-optimization cycle.

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# Part II. The Context Layer

## 5. The Context Layer: From Metadata to Meaning

### 5.1 Anatomy: primitives, models, and patterns

The technologies grouped under "AI data" labels — RAG, vector databases, knowledge graphs, ontologies, GraphRAG — occupy different levels of abstraction, and conflating them produces category errors. The stack decomposes into three kinds of thing:

- **Storage primitives.** *Vector indexes* store embeddings for similarity search; *graph databases* store entities and relationships as first-class citizens, optimized for traversal. Only the latter is a database in the traditional sense.
- **Semantic models.** An *ontology* is a declarative definition of business concepts, their relationships, constraints, and — in its strongest form — permitted actions. It stores no data; it stores meaning. A *knowledge graph* is an ontology populated with instances.
- **Retrieval patterns.** *RAG* retrieves similar chunks from a vector index at query time; *GraphRAG* retrieves connected knowledge via entity relationships and graph traversal, materially outperforming similarity-only retrieval on multi-hop reasoning, entity resolution, and cross-document synthesis [26].

The vertical dependency runs: ontology (meaning) → graph/vector stores (storage) → RAG/GraphRAG (runtime retrieval) [27]. The rise of this entire stack has one cause: the binding constraint on agent quality moved from model capability to context quality. Open protocols solved connectivity — MCP standardizes how agents reach data sources — but a connected agent still must know *what* to retrieve; connectivity without a semantic retrieval layer has been aptly compared to opening every valve to the data lake with no map for the flood [23].

One further piece of anatomy, visible only through the modality lens of Section 1.4, clarifies the competitive map that follows. The contenders for the semantic layer descend from two different traditions. Metric views, semantic views, and the dbt lineage are the *quantitative* semantic tradition — the BI world's discipline of defining measures consistently over numbers. Ontologies, knowledge graphs, and GraphRAG are the *qualitative* semantic tradition — the knowledge-management world's discipline of modeling entities, relationships, and meaning expressed in language. For decades these traditions had separate buyers, separate vendors, and separate conferences. The agent has fused their function: an agent answering one operational question needs governed metric definitions and governed conceptual knowledge *in the same context window*. The land grab of Section 5.3 is, at bottom, the collision of these two lineages onto a single prize — and the same lens explains the hyperscaler contest of Chapter 9, where Microsoft enters from the world's largest qualitative corpus and the data platforms from the quantitative one.

### 5.2 The absorption thesis: vector absorbed, graph contested

The 2023 prediction that standalone vector databases would form a durable category was falsified quickly. Vector search migrated into every incumbent engine — pgvector in Postgres, Atlas Vector Search in MongoDB, native vector types in Snowflake, Databricks, BigQuery, and Oracle 26ai — because a vector index is a feature whose value depends on co-location with governed, fresh, transactional data. The lesson generalizes: *primitives that must sit next to the data are won by whoever owns the data.*

Graph is facing the same absorption pressure (Oracle 26ai's convergence of graph into the single engine is the clearest case [4]) but is resisting it better, for two reasons. First, graph is a data model rather than an index — it carries a query language, a traversal engine, and a modeling methodology, and emulated graph layers over relational engines degrade sharply on deep multi-hop traversal. Second, GraphRAG demand is giving native graph platforms a second commercial life. The sober counterweight is cost: GraphRAG typically runs a multiple of standard RAG cost once graph construction and maintenance are included, which is why hybrid architectures — vector retrieval for breadth, graph traversal for structured multi-hop reasoning, SQL joins for exact figures — have become the production standard rather than wholesale graph adoption [25][26].

### 5.3 The semantic layer land grab

With storage neutral and primitives absorbed, the contested high ground is the semantic layer: the governed definition of what enterprise data means. Three camps are converging on it from different directions, and the distinction between *descriptive* semantics (what does "revenue" mean) and *normative* semantics (what actions are permitted on which entities, by whom) is the most useful lens for comparing them.

**Platform catalogs, building upward.** Databricks shipped Unity Catalog Business Semantics to general availability in April 2026 and open-sourced it: metric views as reusable, SQL- and API-addressable definitions that inherit Unity Catalog governance automatically, plus agent metadata — synonyms, display names, formatting rules — that lets AI tools interpret data in business terms [13][14][15]. Snowflake answered at Summit 2026 with a two-part architecture: Horizon Context, a Collect–Enrich–Activate semantic and governance layer that ingests metadata from external systems, enriches it with lineage and AI-generated documentation, and *activates* it at agent query time rather than merely indexing it; and Cortex Sense, a runtime layer that assembles data, business definitions, and operational knowledge dynamically for agents, with Snowflake claiming a step-change in structured-question accuracy when full business context is supplied versus a frontier model alone [6][7][9][12]. Google evolved Dataplex into Knowledge Catalog, a "universal context engine" that uses Gemini to extract entities and relationships from unstructured files and learns business meaning from usage logs and profiling [36][37][38]. The common platform pattern: target both the governance buyer and the agent buyer, and weaponize automation against the manual-modeling cost that historically gated the category [12].

**Independents, betting on neutrality.** The Fivetran + dbt Labs merger closed on June 1, 2026, creating a combined company of roughly $600M ARR serving over 100,000 data teams, and its flagship joint launch is Agents Schema: an open-source standard that designates a schema in any warehouse as a shared context layer for agents — semantic models, metrics, lineage, and

business documentation stored in ordinary SQL tables, compatible with any warehouse and any SQL-capable agent, operating inside existing governance [16][17][19]. The independent camp's thesis rests on the multi-platform reality of enterprise estates: the large majority of organizations want fewer data-management products but very few want a single vendor [18], and platform semantics stop at the platform boundary. Adjacent to the metric-layer independents sit the knowledge-graph platforms (Stardog, Graphwise, Galaxy, Fluree, and Neo4j's ecosystem), which differentiate on entity-relationship modeling depth and explicitly position themselves as non-invasive semantic infrastructure layered over existing systems — in deliberate contrast to full operational platforms with write-back workflows and tighter platform control [21][22][24].

**The ontology maximalist.** Palantir's ontology is the normative end of the spectrum: by the company's own description, it integrates the elements of a decision — data, logic, and actions — into a foundational representation of the organization, going well beyond the traditional mapping of data to context [20]. For the purposes of this report, the relevant point is comparative: Palantir demonstrates what the *ceiling* of the semantic layer looks like (meaning coupled to permitted action), and thereby defines the gap every catalog-up entrant has yet to close. Descriptive semantics — metric definitions, glossaries, entity maps — can increasingly be reverse-engineered from existing SQL and BI assets. Normative semantics encode organizational tacit knowledge that exists in no codebase, which is why they remain labor-intensive to build and durable once built. A full competitive treatment of operational platforms lies beyond the scope of this report; the conclusion that matters for the data layer is that the semantic layer's value concentrates at its normative end, and every vendor in this chapter is climbing toward it.

### 5.4 Auto-generated semantics: the collapse of the manual-modeling barrier

The traditional barrier to semantic-layer adoption was months of manual ontology and metric modeling [21]. That barrier is now under direct assault from LLM-powered generation: Snowflake's Semantic View Autopilot generates semantic views from existing SQL, Tableau, and Power BI assets; Semantic Studio provides AI-assisted authoring; Cortex Sense assembles context at runtime rather than from a pre-built cache [7][8][9]; Google's Knowledge Catalog infers meaning continuously from usage telemetry [37]. If auto-generation delivers production-grade quality on the messy, undocumented data estates of real enterprises — an unproven claim, as vendor benchmarks predate contact with such estates [12] — the consequences are twofold: the semantic layer's TAM expands dramatically as its marginal cost collapses, and the scarcity premium attached to manually accumulated semantic assets erodes from the descriptive end upward. Our working view: automation commoditizes the bottom of the semantic stack (metrics, glossaries) within two to three years, while the normative top (operational meaning and permitted action) remains human-and-agent co-authored for considerably longer.

### 5.5 Semantic lock-in as the successor to data lock-in

The strategic endpoint of the land grab is already visible to enterprise buyers: embedding business semantics, workflow intelligence, and agent skills in one vendor's orchestration layer creates switching costs of a new kind, and industry observers now explicitly warn that semantic lock-in may become as strategically important as data lock-in once was [11]. The analogy to the table-format wars is exact and is developed in Chapter 10: open semantic

standards are to Unity Catalog and Horizon Context what Iceberg was to proprietary storage — a commoditization threat from below that the leaders are attempting to co-opt by open-sourcing the format while monetizing the runtime.

### 5.6 Compiled context: LLM wikis, Markdown knowledge bases, and OKF

Everything in this chapter so far treats context as something *retrieved*: an agent asks, and infrastructure searches — an index, a graph, a catalog. A fourth pattern, which rose bottom-up from developer practice over the past year, treats context as something *written*: use the LLM once to distill raw documents into a curated, cross-linked knowledge base — typically a directory of plain Markdown files — and let agents simply read it. The business intuition is the difference between an archive and a handbook. Retrieval-based approaches send a brilliant new hire into the file room for every question, re-deriving meaning from the raw pile each time. Compiled context writes down what the organization knows — once, well — and delegates the upkeep. The reason corporate wikis historically failed is that humans will not do the maintenance: cross-references rot, pages go stale, and the wiki is quietly abandoned. The insight popularized by Andrej Karpathy's "LLM wiki" pattern is that this failure condition has been removed — the bookkeeping drudgery that defeats human wiki-keepers (updating every cross-reference, touching a dozen files consistently, never getting bored) is precisely what LLMs do tirelessly — an observation now cited in the format's own genealogy [49][50]. Knowledge is compiled into a versioned corpus and refreshed incrementally as source systems change — the bookkeeping burden shifts from humans to the model, though freshness still depends on reliable synchronization, validation, and review — and every new source enriches the existing graph instead of enlarging an unread pile.

The pattern's field evidence accumulated first in developer tooling — AGENTS.md and CLAUDE.md convention files that coding agents consult before acting, Obsidian-style linked vaults wired into agents, "metadata as code" repositories, the llms.txt convention for agent-facing websites [49]. Because ordinary Markdown links function as graph edges, a well-kept vault *is* a lightweight knowledge graph — which is why this pattern belongs in this chapter and not merely in an engineering appendix.

In June 2026 the pattern crossed from folk practice to published standard. Google released the Open Knowledge Format (OKF) v0.1: knowledge represented as a directory of Markdown files with YAML frontmatter, one concept per file, the file path as the concept's identity, standard Markdown links turning the directory into a knowledge graph, with optional index files for progressive disclosure and logs for change history — no SDK, no runtime, no registry; readable in any editor, shippable in any Git repository [47][48]. The spec explicitly formalizes the LLM-wiki pattern for interoperability, requires exactly one structured field, and ships with reference tooling — including an enrichment agent that walks a BigQuery dataset and drafts a concept document for every table — while Google's Knowledge Catalog both produces and ingests OKF bundles for agent consumption [47][49][50].

Analytically, compiled context does something none of the patterns in Section 5.1 does: it collapses the three-layer stack into a single artifact. The Markdown corpus is simultaneously the store (files), the semantic model (curated concepts and typed-by-convention links), and the retrieval surface (agents read, grep, and navigate it directly). Versus RAG, the trade is compile-time versus query-time: retrieval re-derives meaning from raw chunks on every query; a

compiled bundle is meaning derived once, versioned, and read directly — cheaper per query, dramatically more auditable (every change is a Git diff a human can review), at the cost of being a second copy that must be kept synchronized with its sources [50]. Governance, notably, ports out of the data platform and into the software toolchain — version control, code review, pull requests — which is both the pattern's strength (auditability, total portability) and its present gap (no row-level policy, consistency drift at scale).

The strategic reading admits two endgames, and they map directly onto this report's central tension. In the first, compiled context is the **shadow semantic layer**: plain files plus Git plus MCP is the most radically vendor-neutral context stack conceivable — the practitioner-led antithesis of the semantic lock-in described in Section 5.5, and the federated scenario of Section 10.3 acquiring its most concrete working artifact. In the second, the platforms absorb the pattern and the governed semantic layer becomes a *compiler*, with Markdown bundles as its object code — governed definitions compiled outward into agent-readable files. Google's own architecture already implements the second scenario: Knowledge Catalog serves compiled OKF bundles to agents from a governed core [47][49] — the Iceberg playbook (open the format, monetize the runtime) applied to knowledge itself, within days of the format's birth. We treat the evidence base honestly: this is a practitioner-led movement that was pre-institutional until weeks before this report's publication. But that is precisely the datum — a hyperscaler standardized a developer folk practice within roughly a year of its emergence, which measures how quickly agent-native conventions are becoming the industry's formal architecture, and why a forward-looking reading of this market must watch developer practice as closely as vendor roadmaps.

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# Part III. Reasoning and the Closing Loop

## 6. The Analytical Reasoning Layer: The Data Layer Learns to Decide

### 6.1 The dual shift: frequency revolution and the causal-ladder ascent

The classical economics of analytics rationed sophistication. Descriptive statistics and BI — rung one of Pearl's causal ladder, association — were cheap and automated; predictive modeling — statistically sophisticated, but still rung-one association — cost data-scientist-weeks; causal analysis (rung two, intervention) cost econometrician-months; what-if simulation (rung three, counterfactual) was a consulting engagement. The vast majority of enterprise decisions were therefore made on rung-one information plus intuition, with upper-ladder analysis rationed to a handful of large decisions per year.

Agentic AI breaks the rationing in two directions simultaneously. The *frequency revolution* is a Jevons effect: when the unit cost of an analysis collapses from human-weeks to agent-minutes, demand explodes — demand forecasting moves from quarterly to per-SKU-daily, price-elasticity analysis from annual to per-campaign. The *ladder ascent* is structural: an agent that acts cannot stop at association. Knowing that revenue fell (observation) does not select an action; the agent needs intervention estimates (what happens if we cut price 5%?) and counterfactuals (what would have happened had we not?). Acting AI has causal and simulation demand built

into its job description. Industry analysis has begun formalizing the consequence as a new layer of the AI stack — decision intelligence — enabling agents to test interventions, run counterfactual scenarios, and produce decision-grade outputs that are explainable and auditable [29].

### 6.2 Predictive analytics: the fastest-commoditizing layer

Prediction is the layer absorbing into platforms fastest. Agentic data-science tooling is generally available across the majors — Google's Data Engineering Agent and Data Science Agent reached GA in 2025–2026, automating pipeline construction and model development inside BigQuery [38][39] — and time-series foundation models (Google's TimesFM, Amazon's Chronos lineage) increasingly let agents skip bespoke model construction altogether. The investment implication mirrors Section 5.2's absorption logic: predictive *modeling* ceases to be a value-capture point, and value migrates to (a) the event-level data quality and freshness that feed predictions and (b) the metered compute predictions consume. Prediction's commoditization is a margin threat to analytics-services businesses and a volume tailwind to consumption-priced data platforms.

### 6.3 Causal analytics: the trust wall and the experimentation flywheel

Causality is where the reasoning layer becomes strategically interesting, because it is where the dominant agent stack hits a wall. LLMs are correlation-native; chained reasoning and retrieval produce fluent explanations, but fluency does not scale into accuracy, explainability, or auditability — a coherent-sounding explanation is not a defensible decision, and without causal machinery agents struggle to isolate true drivers from confounders or justify one action over another as conditions change [29]. Enterprise recognition is moving fast: survey work by Dataiku and Databricks, fielded in 2025, found only 16% of organizations using causal AI but roughly 70% planning adoption by 2026 [28], and analysts have begun calling 2026 the breakout year for causal decision intelligence, with a specialist vendor ecosystem (more than twenty companies under coverage at theCUBE Research) forming around it [28][29]. Third-party market sizings for causal AI imply forty-plus percent compound growth through 2034 [30]; we treat such figures as directionally indicative at best, but the direction is unambiguous.

Three structural observations matter for the data layer:

**The ontology is the natural home of the causal graph.** Causal inference presupposes causal structure — a directed graph of what can influence what — and that structure cannot generally be identified from observational correlations alone; extracting it requires additional assumptions, domain knowledge, temporal ordering, or intervention data, which is why in practice it is codified business understanding. This is the same species of asset as normative semantics (Section 5.3): the semantic layer's next evolution is from defining metrics to encoding the causal relationships *between* metrics, and the vendors contesting the semantic layer are therefore, knowingly or not, contesting the causal layer too.

**Agent actions are intervention data.** The gold standard of causal knowledge is intervention, and acting agents mass-produce interventions: every price change, campaign launch, and inventory reallocation is — if logged with context and, ideally, randomized at the margin — an experiment. This closes a flywheel: act → accumulate intervention data → improve causal models → act better. Organizations that close the loop compound an advantage; organizations that do not accumulate fluent, confounded beliefs at machine speed. OpenAI's 2025 acquisition of

the experimentation platform Statsig signaled that frontier labs understand experimentation infrastructure as a determinant of how fast agentic systems learn; we expect data platforms to reach the same conclusion via M&A (Section 11.2).

**Intervention logs become a strategic data class.** Causal analysis is most reliable when it has unit- or event-level data together with a record of what was done, to whom, when, and under what assignment mechanism. Aggregate data can support some designs — synthetic control, grouped difference-in-differences — but it sharply limits identification, heterogeneity analysis, and confounder adjustment. This elevates a previously mundane artifact — the intervention log — into core data-layer infrastructure, alongside the decision provenance records discussed in Section 6.6: the data layer should preserve intervention history at event granularity wherever possible.

### 6.4 Inference governance: hallucinated causality at machine speed

The frequency revolution has a failure mode that deserves its own name. Agents generate plausible analyses — complete with causal narrative — faster than human organizations can audit them, and statistically invalid comparisons (selection bias, survivorship, post-hoc rationalization) arrive dressed in fluent prose. We call the risk *hallucinated causality*, and the mitigation is a third governance discipline alongside the two the industry already practices — *read governance*, controlling which data humans and agents may access, and *action governance*, controlling which actions agents may execute (the perimeter contested in Sections 5.3 and 8.2): **inference governance** — encoding a "grammar of valid comparison" into the semantic layer itself, specifying which metrics may be causally compared, which experiment designs are required for which claims, and which analytical conclusions are decision-grade versus exploratory. No vendor ships this today in mature form; it is, in our view, among the most predictable product categories of 2027, and the semantic-layer incumbents are best positioned to host it.

### 6.5 What-if simulation: the enterprise world model

Counterfactual reasoning — rung three — is the most expensive and most valuable layer, because evaluating an action without taking it requires a *world model* of the business. This converges with the industry's "system of intelligence" framing: a deterministic and cognitive digital twin that understands business state, captures rules and exceptions, supports human judgment, and teaches agents to improve [34][35]. The academic literature has begun formalizing the composition — agentic AI plus digital twins plus model-based optimization and simulation — as a distinct architecture for operations and supply-chain decision support [33], and commercial category formation is visible at the high end: Alembic's 2026 platform release runs real-time enterprise causal simulation on dedicated NVIDIA Grace Blackwell supercomputing infrastructure, marketed as letting companies simulate the financial impact of strategic decisions before committing capital [31].

Simulation forces three requirements onto the data layer:

**Branching becomes a first-class primitive.** A what-if run executes candidate actions against a copy of current state. Copy-on-write branching — Neon's signature capability, now core to Lakebase, mirrored by Iceberg's branch-and-tag semantics — is the primitive that makes this affordable, and simulation industrializes it: one decision = N candidate actions × M scenarios = N×M ephemeral branched environments [2]. "Git for data" graduates from developer con-

venience to load-bearing infrastructure, structurally favoring serverless, storage-separated architectures.

**State synchronization is the quality bottleneck.** A simulation is only as good as its twin's fidelity, and the central deployment obstacle documented for enterprise digital twins is the absence of a unified data fabric: when telemetry is trapped in fragmented regional databases and legacy silos, the virtual representation drifts from physical state, and simulations fail or mislead [32]. The dependency is vertical and strict — the reasoning layer's quality reduces to the convergence agenda of Part I. Simulation is thus not merely a beneficiary of data-layer unification; it is a forcing function for it.

**Simulation compute is a new demand class.** Counterfactual analysis at agentic frequency creates compute demand distinct from model inference — Monte Carlo over branched state, agent-based simulation, optimization solvers. That a causal-simulation specialist runs on DGX-class infrastructure [31] is an early marker: the "business twin" may become to enterprise data platforms what the physical twin (Omniverse) is to industrial NVIDIA — a second simulation market, metered by the platforms that host the state being simulated.

### 6.6 New strategic data assets: the next systems of record

The reasoning layer mints new data categories whose strategic weight is not yet priced into vendor analysis: *intervention logs* (Section 6.3); *decision provenance* — the linked record of what analysis led to what action with what expected and realized outcome, which is simultaneously the audit trail regulators will demand and the training corpus for organizational learning loops; and *agent memory* at the organizational rather than session level — institutional memory as a queryable asset, which in practice increasingly takes compiled-context form: curated Markdown knowledge the agent itself maintains (Section 5.6). Whoever owns these stores owns the learning loop, and the learning loop is the closest thing to a structural moat the agentic stack offers. The contest for them is already visible in product design: agent memory embedded in the database engine (Oracle) versus the managed agent runtime (AWS AgentCore) versus the data platform (Lakebase-adjacent state stores) [4][42].

### 6.7 The decision-loop compute multiplier

The economics of the reasoning layer compound the consumption thesis of Section 4.5 by an order of magnitude. A single agentic decision implies a fan-out: several predictive queries to characterize the situation, causal queries to identify drivers, and N×M simulation branches to evaluate candidate actions — each leg metered. Where the dashboard era priced one query per human question, the agentic era prices a *loop* per decision and runs the loop continuously. This multiplier is, in our assessment, the most underappreciated long-duration driver of data-platform consumption growth — and, symmetrically, the strongest argument that agent FinOps (Section 4.5) graduates from nuisance to board-level cost program within two budget cycles.

## 7. Closing the Loop: From Analysis Back to Action

The classical data layer was a one-way street. Operational systems recorded the business; pipelines carried the records into warehouses; analysis produced insight — and there the architecture stopped. The last mile, from insight back to the systems that actually run the business, was closed by hand: a human read the dashboard, decided in a meeting, and re-keyed the conclusion into an ERP transaction, a payroll adjustment, a banking instruction.

The loop existed, but it ran at human latency, leaked at every manual step, and preserved no record connecting evidence to action to outcome.

The architecture assembled across this report points toward the completion of that loop in the infrastructure itself. The reasoning layer of Chapter 6 must not terminate in a report: its outputs — a price change, a reallocation, a journal entry, a credit decision — must flow back, under governance, into the transactional systems that manage the business's actions: ERP, payroll, banking, CRM, supply-chain execution. Analysis that ends at OLAP is advice; analysis that writes back into OLTP is management. This is the principle Palantir turned into a product philosophy — data, logic, and action held in one representation [20] — and the converged data layer's trajectory is to generalize it.

The two convergences of Part I are what make the completed loop coherent for the first time. With transactional and analytical engines sharing one platform, one catalog, and one governance perimeter (Chapter 3), a decision's evidence, its simulations, its execution, and its measured outcome can inhabit a single governed system — decision provenance (Section 6.6) supplying the thread that ties them together, and the experimentation flywheel (Section 6.3) supplying the return path from outcome to better model. Each development chronicled in this report — transactional engines acquired into analytical platforms, catalogs extended to actions and agent identity, context activated at query time — is a segment of this loop under construction.

Completion will be sequenced by consequence. Write access to regulated systems of record — ERP, payroll, core banking — is granted politically and audited heavily, so the loop will close first around low-consequence actions and last around the ledger. But the direction is unambiguous, and it furnishes the summary image of the architecture assembled across Parts I and II: every layer examined here — unified storage, transactional integration, governed context, causal machinery, simulation, provenance — is a component of a single closed loop: **sense → understand → reason → act → learn**. The data layer's destination is not a better warehouse. It is the substrate of that loop.

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# Part IV. Competitive Landscape

## 8. The Converged-Market Contenders: Strategy Profiles

### 8.1 Databricks: lakehouse to full stack, openness as strategy

Databricks holds the broadest quadrant coverage of any independent. From its unstructured-analytics home it took structured analytics (Photon, SQL warehousing), then transactional workloads (Neon → Lakebase, GA February 2026), uniting analytics, ML, and a native managed Postgres layer under one governance perimeter — a genuine full-stack data platform rather than an analytics environment [2]. Its strategic signature is the recurring open-format playbook: open-source the layer below (Delta interoperability, Iceberg REST support, and in April 2026 the open-sourcing of Unity Catalog Business Semantics itself) and monetize the governed runtime above — Unity Catalog as the control point through which every engine, metric, and agent passes [1][13][15]. The structural weakness is constituency: Databricks' gravity

is strongest with engineering organizations, weakest with the business-user surfaces (BI, finance, operations) where Microsoft and Snowflake distribute — a gap the Genie/Databricks One push is designed to close, so far only partially.

### 8.2 Snowflake: warehouse to context platform

Snowflake's repositioning has been the most aggressive narrative arc of the convergence era: from "the data cloud" to the context and governance substrate of the agentic enterprise. The June 2026 Summit consolidated the architecture — Horizon Catalog as the governance spine; Horizon Context as the collect-enrich-activate semantic layer, with reference customers using it to give AI a shared definition of enterprise truth; Cortex Sense as runtime context assembly; semantic-view autogeneration to collapse the modeling barrier; the agent surface renamed CoWork and repositioned from answers to actions [10][11]; Cortex Code extended beyond Snowsight through an SDK, MCP server, and editor integrations, carrying the governed agent stack out to developer surfaces [51]; enterprise MCP connectivity and governance — including agent identity — added through the pending Natoma acquisition, which extends Snowflake's governance perimeter from data assets to agent actions [45]; and Postgres (via Crunchy Data) supplying the transactional leg [6][7][9][12]. Two tensions define the risk profile. First, proof: the headline accuracy and automation claims are preview-stage, untested against genuinely messy enterprise estates [12]. Second, the consumption model cuts both ways — the decision-loop multiplier (Section 6.7) accrues to Snowflake powerfully, but agent-driven query storms make Snowflake the most exposed major to the eventual agent-FinOps optimization cycle (Section 4.5).

### 8.3 Oracle: the converged-engine defense

Oracle's 26ai strategy is the structural outlier — single-engine fusion of vector, JSON, graph, and relational with in-engine agent memory and cell-level security bound to agent identity, positioning the database itself as the center of gravity for enterprise agentic workloads [3][4]. Its assets are real: the deepest installed base of mission-critical systems of record, mature write-governance machinery (approval workflows, segregation of duties) that the analytics-born platforms must build from scratch, and a vertically integrated application estate. Its exposure is equally structural: the agent-native flow (Section 4.3) is being won on Postgres-compatible, developer-led surfaces, and AI-assisted migration tooling is lowering the historical exit costs that protected the stock — observers note that the large pool of spend tied up in legacy relational databases is becoming more contestable as migration friction falls [43]. Oracle's franchise is best modeled as a slowly leaking reservoir with a strong new dam (26ai) whose effectiveness depends on convincing agent builders to build *inside* the engine — a developer-mindshare battle Oracle enters from behind.

### 8.4 MongoDB: AI-native re-founding of the document quadrant

MongoDB chose depth over breadth: rather than contest analytics, it is re-founding its home quadrant as the default store for AI-native applications — Atlas Vector Search with aggressive cost engineering (binary quantization), stream processing to collapse adjacent architecture, an MCP server letting agents natively understand schemas and query autonomously, and embedding-model assets via the Voyage AI acquisition [43][44]. The logic is sound: agent-generated data is unstructured and schema-fluid, and the document model is its natural container; Atlas growth (roughly 29% in mid-FY2026, three-quarters of revenue) shows the strategy con-

verting [44]. Two structural threats bound the upside: open-source standardization pressure on the document model itself (the Open DocumentDB project) and hyperscaler competition [43]; and — more fundamental — Postgres encroachment, as pgvector-plus-JSONB inside Lakebase-class platforms offers "good-enough MongoDB plus good-enough vector" inside a governed analytical estate, attacking MongoDB's quadrant from the flank.

### 8.5 PostgreSQL as battleground

PostgreSQL deserves treatment as terrain rather than vendor. Because the project is a commons, the structured-transactional quadrant's future is contested at the *commercialization chokepoints* — managed serverless runtimes, branching infrastructure, extensions, and platform integration — rather than at the engine. The 2025 wave (Neon to Databricks, Crunchy Data to Snowflake, Microsoft building HorizonDB) consolidated the most strategic chokepoints, but the commons keeps regenerating candidates: the remaining independent serverless-Postgres and Postgres-platform assets (Supabase the most prominent, alongside infrastructure specialists in the branching/storage niche) are the obvious residual targets for any platform — hyperscaler or data major — that missed the first wave. The meta-point for investors: in a commons-based quadrant, M&A is the share-shift mechanism, and the acquisition premium migrates to whichever chokepoint the agent workload makes scarce next (our candidate: branching-at-scale infrastructure, per Section 6.5).

## 9. Hyperscaler Data Platforms

### 9.1 Microsoft: distribution-led integration

Microsoft Fabric is the bundling strategy in its purest form: OneLake as a single logical lake, with warehousing, real-time analytics, BI, and data science delivered as SaaS, fused to the M365/Copilot distribution machine and now extended downward by HorizonDB's transactional leg [5]. Fabric rarely wins on best-of-breed depth; it wins on default presence — "good enough, already included, Copilot-attached" — the classic Microsoft mid-market capture playbook. The modality lens of Section 1.4 identifies the deeper asset beneath the bundle: M365 gives Microsoft custody of the world's largest governed qualitative corpus — documents, mail, meetings — while Fabric and HorizonDB annex the quantitative estate, making Microsoft the one contender entering the convergence with distribution over both modalities. For the independent platforms, Fabric is less a feature competitor than a procurement competitor: it attacks at renewal time, through the CFO.

### 9.2 Google: the Agentic Data Cloud

Google's Cloud Next 2026 release was the most coherent hyperscaler statement of the converged thesis: an Agentic Data Cloud organized around delivering cleaned, organized, governed context to agents [37]. Knowledge Catalog (the Dataplex evolution) supplies the semantic foundation, using Gemini to extract entities and relationships from unstructured assets and learning enterprise meaning from usage telemetry; the Gemini Enterprise Agent Platform absorbs Vertex AI as the home for building and governing agents; data-engineering and data-science agents are GA inside BigQuery [36][37][38][39][40]. The strategically distinctive move is *zero-copy federation* with partners including Palantir, Salesforce, and Workday, letting agents query data resident on other clouds without movement — positioning BigQuery as the cross-cloud reasoning engine rather than demanding data centralization [36]. Where Microsoft gates with distribution, Google bids with technical coherence plus model and TPU economics; its

persistent gap is the enterprise work surface, where Workspace's footprint trails M365 — in the modality terms of Section 1.4, a structurally smaller governed qualitative corpus.

### 9.3 AWS: the neutrality strategy

AWS is running a different play. Its convergence assets are real — SageMaker Unified Studio fuses EMR, Glue, Athena, Redshift, Bedrock, and SageMaker AI into one governed, Iceberg-based workspace [41][46], and Bedrock AgentCore has matured into a full operational layer for stateful agents, including managed memory, evaluation tooling, and even experimental agent-payment infrastructure [42]. What AWS conspicuously declines to build is an opinionated business-semantics layer or a flagship work surface; its position is the neutral substrate on which Databricks, Snowflake, Palantir, and the frontier labs all run. The strategy preserves optionality and Capex leverage, but carries a structural dilution risk: as agentic value capture migrates up-stack toward context and reasoning (Part II), the neutral-substrate share of each enterprise dollar shrinks relative to the opinionated platforms' share.

## 10. Standards and Interoperability Wars

### 10.1 Table format endgame

The Iceberg war is effectively settled as a draw that everyone claims to have won: Iceberg REST Catalog support is table stakes, both leaders ship interoperable managed tables, and bi-directional cross-engine access through governed catalogs (Unity Catalog; Horizon's Polaris-powered catalog) is generally available [1][9]. The settlement's true beneficiary is the catalog layer itself — neutral storage made the catalog the chokepoint — and its true casualty is any strategy premised on storage gravity alone. The format endgame is therefore best read as the template for the semantic wars now beginning.

### 10.2 Semantic standards: the Iceberg analogy

Three open bets are live. Databricks open-sourced Unity Catalog Business Semantics with an Apache Spark implementation, making metric definitions portable by design [13][15]; Fivetran + dbt launched Agents Schema as a warehouse-agnostic open standard for agent context [16] [17]; and the Open Semantic Interchange (OSI), convened by Snowflake in September 2025 with dbt Labs, Salesforce, RelationalAI, and a coalition of BI and catalog vendors, is drafting a vendor-neutral semantic model specification [53]. The Iceberg analogy implies the likely equilibrium: if a semantic interchange standard achieves critical adoption, the *format* commoditizes and value re-concentrates in semantic *generation, quality, and runtime activation* — favoring vendors playing "open format, proprietary runtime" (Databricks explicitly; Snowflake implicitly via Horizon's activation layer) and pressuring closed-ontology premiums from below. The key watch item is whether Snowflake and Google adopt, embrace-and-extend, or ignore Agents Schema; adoption would mark the semantic layer's Iceberg moment. The June 2026 Data + AI Summit sharpened the test: Databricks extended Unity Catalog Semantics with Glossary — authoritative concepts, terms, and taxonomies, drafted by an agent and linked to the underlying data — and Domains, and declared Metrics OSI-ready, aligning itself with the standard its principal rival convened [52]. That both leaders now claim the same interchange standard while Databricks pushes its own semantics upward into taxonomy is the Iceberg pattern reproduced almost exactly — convergence on a common format, competition displaced onto the runtime above it — and it moves OSI conformance ahead of Agents Schema as the adoption metric worth tracking.

### 10.3 The integration premium question

Every integrated platform's pitch reduces to one claim: data, semantics, and reasoning must co-reside for governance and latency. Open protocols are the standing counterclaim. MCP standardized agent-to-data connectivity; open semantic standards aim to make context itself portable; and if both succeed, an agent could assemble governed context across best-of-breed systems without a mega-platform intermediary — re-fragmenting the market the convergence era is consolidating. The compiled-context pattern (Section 5.6) supplies this scenario with its most concrete working artifact — plain files, Git, MCP, and now an open interchange format in OKF — and enterprise adoption of OKF-class bundles, together with the OSI-conformant semantic interchange that Section 10.2 identifies as the adoption metric to track, is accordingly the sharpest leading indicator for the federated branch. Our base case assigns the integration premium durability through our five-year forecast window, on three grounds: protocol connectivity does not solve context *selection* (the map-for-the-flood problem [23]); cross-system governance remains primitive relative to in-platform inheritance; and the decision-loop's latency economics favor co-location. But the federated scenario is live, asymmetric in payoff (it primarily devalues platform multiples rather than platform revenues), and best monitored through semantic-standard adoption (10.2) and the maturation of cross-system agent governance.

## 11. Specialized and Emerging Segments

### 11.1 Graph databases: real demand, bounded by cost

GraphRAG demand is the graph segment's second act, with hybrid vector-plus-graph architectures the emerging production standard for complex enterprise retrieval and reasoning [25]. The bull case rests on measurable precision advantages for multi-hop and cross-document reasoning and on knowledge graphs' role as the causal-structure substrate (Section 6.3); the discipline comes from cost — multi-x premiums over plain RAG for construction and maintenance [26] — and from absorption pressure as converged engines ship graph features [4]. The investable distinction is between graph as *feature* (absorbed) and graph as *platform for the semantic/causal layer* (defensible); Neo4j and the standards-based platforms are racing to establish the latter identity before the former framing hardens.

### 11.2 Experimentation and causal-AI platforms

The causal decision-intelligence ecosystem — twenty-plus specialists under analyst coverage, spanning causal-inference platforms, vertical reasoning engines, and high-end simulation (Alembic) [28][29][31] — is, in our assessment, a category whose strategic value exceeds its standalone commercial gravity. Its assets (causal modeling machinery, experimentation infrastructure, intervention-data tooling) are precisely what the data platforms lack as they climb the reasoning ladder, and the frontier-lab precedent (OpenAI/Statsig) established the acquisition logic. We expect the segment's exit map to be dominated by platform M&A within two to three years, with independent survival concentrated among vertical specialists whose domain causal models (life sciences, marketing mix) resist generalization.

### 11.3 Agent memory and decision-intelligence infrastructure: category watch

Agent memory is currently being claimed by three architectural camps simultaneously — the database engine (Oracle's in-engine Unified Memory Core [4]), the agent runtime (AWS AgentCore's managed memory [42]), and the data platform (lakebase-adjacent state stores [2])

— while decision provenance (Section 6.6) has no incumbent at all. Category formation here is early enough that the correct posture is monitoring rather than conviction; the signal to watch is whether memory and provenance consolidate into the data platform's governance perimeter (extending the integration premium) or harden into an independent layer (the federated scenario's strongest beachhead).

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# Part V. Conclusion

## 12. The Substrate of the Loop

The argument of this report compresses into three movements. First, the 2x2 world collapsed — along the form axis within analytics, then along the workload axis — because a new class of consumer, the agent, analyzes and acts in a single machine-speed loop that the old fragmentation cannot serve (Part I). Second, with storage neutralized by open formats, value migrated into two new strata: the context layer, which industrializes the newly computable qualitative corpus, and the analytical reasoning layer, which turns prediction, causal inference, and simulation into high-frequency agent workloads (Part II). Third, every contender profiled in Part III is assembling the same object from a different starting point: the closed loop of Chapter 7 — sense, understand, reason, act, learn — held inside one governance perimeter.

What remains open is Thesis 5, and it should be adjudicated on observables rather than narrative: OSI conformance across the major platforms (Section 10.2); enterprise adoption of OKF-class compiled-context bundles (Section 10.3); whether agent memory and decision provenance consolidate into platform governance perimeters or harden into an independent layer (Section 11.3); and the arrival timing of the agent-FinOps optimization cycle (Section 4.5). Our base case holds the integration premium durable through the five-year window, but the federated scenario is live and its payoff asymmetric — it devalues platform multiples before it devalues platform revenues. Either way, the destination is the one Chapter 7 names: the data layer's future is not a better warehouse but the substrate of the loop.

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# Appendices

## Appendix A. Glossary of Convergence-Era Terminology

| Term | Definition |
|---|---|
| **Lakehouse** | Architecture layering warehouse properties (ACID, schema, performance) on open object storage; the post-2020 analytical default. |
| **Lakebase** | A serverless, branching-capable transactional (Postgres) layer native to the lakehouse platform; category formalized with Databricks Lakebase GA (Feb 2026). |

| Term | Definition |
|---|---|
| **Open table format** | Engine-neutral table specification (Apache Iceberg, Delta Lake) enabling multi-engine read/write on shared storage. |
| **Context layer** | The stratum that supplies agents with governed meaning: catalogs, semantic models, ontologies, knowledge graphs, retrieval infrastructure. |
| **Descriptive vs. normative semantics** | Definitions of what data means (metrics, glossaries) vs. encodings of permitted action and operational meaning; the latter is costlier to build and more durable. |
| **Semantic view / metric view** | Governed, reusable definition of business metrics addressable by SQL, BI tools, and agents (Snowflake / Databricks implementations respectively). |
| **Context activation** | Serving semantic context dynamically to agents at query time, rather than as a passive index (e.g., Horizon Context's activate stage; Cortex Sense). |
| **GraphRAG** | Retrieval pattern using entity relationships and graph traversal rather than (or alongside) vector similarity. |
| **Branching ("git for data")** | Copy-on-write forking of database or table state; the enabling primitive for agent experimentation and what-if simulation. |
| **Intervention log** | Event-level record of actions taken (what, to whom, when, under what assignment), the raw material of causal inference. |
| **Decision provenance** | Linked record connecting analysis → action → expected and realized outcome; audit trail and learning-loop substrate. |
| **Inference governance** | Governance of analytical conclusions: encoded rules for valid comparison, required experiment designs, decision-grade thresholds; the third discipline alongside read governance (data access) and action governance (permitted agent actions). |
| **Agent FinOps** | Cost governance for agent-driven consumption: budget routing, per-session ceilings, cost-aware query planning. |
| **Semantic lock-in** | Switching costs arising from embedding business meaning in one vendor's semantic/orchestration layer; successor to data lock-in. |
| **Modality axis** | Classification of data by natural processor: *quantitative* (reducible to numbers/codes; computable by deterministic code and SQL, including categorical values) vs. *qualitative* (meaning resisting such reduction; natively and generally computable only by probabilistic models, i.e., LLMs — earlier NLP extracted narrow, predefined signals via task-specific pipelines). Orthogonal to the structured/unstructured form axis. |

| Term | Definition |
|---|---|
| **Dark data** | Stored-but-uncomputed enterprise information — predominantly the qualitative corpus (documents, mail, decks) — revalued by LLMs from storage cost to computable asset. |
| **Compiled context** | Context supplied by pre-compiling knowledge into a curated, versioned, agent-readable corpus (typically Markdown), rather than retrieving it from raw sources at query time. |
| **LLM wiki** | Practitioner pattern in which an LLM builds and maintains a cross-linked Markdown knowledge base — the maintenance labor that defeats human wikis delegated to the model. |
| **OKF (Open Knowledge Format)** | Google's open v0.1 specification (June 2026) formalizing the LLM-wiki pattern: knowledge as a directory of Markdown files with YAML frontmatter, links forming a knowledge graph; "format, not platform." |
| **Shadow semantic layer** | Bottom-up, file-based semantic assets (compiled context in Git) accumulating outside the governed platform semantic layer. |

## Appendix B. Agentic Data Workload Taxonomy (Reference)

| Workload class | Access pattern | Natural substrate | Primary contenders |
|---|---|---|---|
| Agent episodic memory | High-frequency transactional read/write | Serverless OLTP | Lakebase, Snowflake Postgres, AgentCore Memory, Oracle 26ai |
| Agent semantic memory | Graph/entity read-heavy | Knowledge graph / converged engine | Neo4j-class, Oracle 26ai, platform catalogs |
| Session state & checkpoints | Low-latency durable writes | Serverless OLTP | Lakebase-class |
| Experimental branches | Ephemeral copy-on-write forks | Branching storage | Neon/Lakebase, Iceberg branches |
| Embedding pipelines | Batch + streaming inference | Lakehouse compute | Databricks, Snowflake Cortex, BigQuery |
| Entity/relationship extraction | LLM-powered batch | Lakehouse + catalog | Knowledge Catalog, Unity Catalog, Horizon Context |
| Predictive workloads | Scheduled + on-demand compute | Warehouse ML / foundation models | All platforms; TimesFM/Chronos-class models |

| Workload class | Access pattern | Natural substrate | Primary contenders |
|---|---|---|---|
| Causal / experimentation | Event-level scans + assignment logs | Warehouse + experimentation infra | Platforms + causal specialists |
| What-if simulation | N×M branched compute bursts | Branching + simulation compute | Lakebase-class, specialist simulators, NVIDIA-class infra |
| Decision provenance | Append-heavy, audit-queried | (Unclaimed) | Open |

## Appendix C. Timeline of Consolidation Events (2024–2026)

- **2024** — Databricks acquires Tabular (Iceberg founders); Iceberg/Delta interoperability race begins in earnest.
- **May 2025** — Databricks acquires Neon (serverless Postgres, branching).
- **June 2025** — Snowflake acquires Crunchy Data; Databricks announces Lakebase (public preview) and Unity Catalog Metrics; Unity Catalog adds full Iceberg REST support [1].
- **2025** — OpenAI acquires Statsig (experimentation infrastructure); Microsoft announces Azure HorizonDB (private preview, November) [5].
- **October 2025** — Fivetran and dbt Labs announce all-stock merger [16].
- **February 2026** — Databricks Lakebase reaches GA; "lakebase" category formalized [2].
- **March 2026** — Oracle AI Database 26ai agentic suite (Unified Memory Core, in-engine agent governance) [3][4]; Alembic launches real-time causal enterprise simulation on NVIDIA DGX [31].
- **April 2026** — Databricks open-sources Unity Catalog Business Semantics at GA [13][15]; Google Cloud Next 26: Agentic Data Cloud, Knowledge Catalog, Gemini Enterprise Agent Platform [36][37][38]; Snowflake expands Cortex Code beyond Snowsight through an SDK, MCP server, and editor integrations (April 21) [51].
- **May 2026** — Snowflake announces a definitive agreement to acquire Natoma, an enterprise MCP connectivity and governance platform (May 27; pending close), extending its governance perimeter from data assets to agent actions [45].
- **June 2026** — Fivetran + dbt merger closes; Agents Schema launched [16][17]. Azure HorizonDB enters public preview (June 2) [5]. Snowflake Summit 26: Horizon Context, Cortex Sense, CoWork action pivot [6][7][9][11][12]. Databricks Data + AI Summit 26: Unity Catalog Semantics extended with Glossary and Domains; Metrics declared OSI-ready [52]. Google publishes Open Knowledge Format (OKF) v0.1, formalizing the LLM-wiki pattern; Knowledge Catalog ingests and serves OKF bundles [47][48][49][50].

## Appendix D. Data and Sourcing Notes

The following claims are vendor-sourced, preliminary, or otherwise flagged, consistent with this series' verification discipline. (1) Snowflake's Cortex Sense accuracy benchmark (86% vs. 24%) and Semantic View Autopilot automation claims are vendor-reported and preview-stage; no independent replication on production enterprise estates exists as of this writing [12]. (2) Databricks Lakebase's February 2026 general availability applied initially to AWS, with

Azure GA following in March 2026; adoption-rate comparisons are company-reported [2]. (3) Azure HorizonDB performance figures (up to 3,072 vCores, 128TB databases, roughly 3x open-source Postgres transactional throughput) are Microsoft benchmarks without independent verification; the service remains in public preview [5]. (4) The Natoma acquisition is a signed definitive agreement pending customary closing conditions as of this writing; financial terms were not disclosed [45]. (5) Causal-AI market sizings [30] and the 2025 Dataiku/Databricks adoption survey figures (16% current use; approximately 70% planned by 2026) [28] are treated as directional indicators only. (6) Neon's pre-acquisition statistic that the majority of new databases on its platform were agent-created is vendor-reported. (7) OKF is a v0.1 draft specification; adoption beyond Google's own tooling is not yet evidenced, and OKF-dependent arguments should be read as conditional on that adoption [47]. (8) Estimates of the qualitative ("dark data") share of enterprise information, commonly cited at 80–90%, are industry rules of thumb rather than measured figures.

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*Forward-looking statements are subject to revision as the underlying evidence develops; vendor-sourced and preliminary claims are identified in the Data and Sourcing Notes.*

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